Silence, expansion speed, and a Dirichlet law for early contact

Mathematical research note

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Abstract

Consider Poisson attempts to originate expanding civilizations in Euclidean space, with independently assigned subluminal speeds and immediate persistent light-speed signatures. Condition on an ideal observer seeing no signature by time T. If the speed gap $q = c/V - 1$ has distribution $F(z) \sim C z^{\alpha}$ near zero, the probability of contact within a further time h has exponent $1 + min(\alpha, d)$, with a logarithmic correction at $\alpha = d$. Conditional on such early contact, the fractions of the horizon spent awaiting the first signal, between first signal and contact, and after contact converge jointly to $Dirichlet(1, min(\alpha, d), 1)$. We derive exact independent-front formulas and an explicit Poisson-shell coupling that transfers the asymptotic results to a process rejecting births inside earlier accepted expansions. The mathematics is proved under stated assumptions and checked computationally. The exact combination was not found in the literature searched, but historical originality is unverified. Several ingredients are known, including Poisson cone calculations, random-speed nucleation, and regular-variation asymptotics. The result does not establish which model describes extraterrestrial life and does not resolve the empirical Fermi paradox.

Outcome and research scope

The strongest result of this investigation is a theorem about the conditional geometry of early contact after ideal silence. It identifies both a change of exponent and a limiting distribution, and it remains valid after a physically motivated correction to the simplest independent-front model.

The investigation did *not* decisively solve the Fermi paradox. A statement about alien abundance, behavior, or technology requires evidence connecting model assumptions to our universe. A mathematical theorem cannot supply that missing evidence.

Three substantially different directions were explored before selecting this one:

1. **Graph growth and spatial inverse problems.** Conditional unoccupied-site probabilities lead to distance kernels and, on a ray, a dyadic inverse transform. Standard distance embeddings, percolation, and inverse nucleation results make foundational novelty uncertain.
2. **Observer selection and statistical identifiability.** Two-point latent-rate constructions give exponentially hard inference of a population mean from surviving observers. The construction is rigorous but close to established frailty and nonparametric impossibility methods.
3. **Causal geometry with heterogeneous speeds.** Exact finite-age formulas expose a divergence if the age boundary is discarded. Keeping that boundary produces the speed-tail transition below. A separate coupling overcomes the random-speed phantom-birth problem.

The third direction was selected because it yields a complete, falsifiable model theorem with explicit constants, a joint limit law, and a quantified robustness estimate. It also identifies a precise limitation of what the chosen silence datum can infer.

Definitions and assumptions

Fix $d \in \{1,2,\ldots\}$, $c>0$, $\lambda>0$, and $T>0$. Write

$$v_d = vol\{x \in \mathbb{R}^d : |x| \le 1\}.$$

Potential births form a Poisson point process on $\mathbb{R}^d \times [0,\infty)$ with intensity $\lambda\, dx\, dt$. Each receives an independent speed $V \in (0,c)$. Let

$$q = \frac{c}{V} - 1 > 0, F(z) = P(q \le z).$$

No density for F is required. The observation point is fixed in advance at the origin; it is not sampled from a population of surviving observers.

For a potential birth (x,t,V), define its hypothetical signal-arrival and expansion-arrival times at the observer by

$$D = t + \frac{|x|}{c}, A = t + \frac{|x|}{V}.$$

We compare two precisely specified processes:

- **Free process:** every potential birth emits a signature immediately and its ball expands forever at speed V.
- **Accepted process:** a potential birth is rejected if its birthplace is already covered by an earlier accepted expanding ball. Every accepted birth emits a signature immediately and its ball continues growing at its assigned speed. Overlap does not halt an accepted ball.

Thus this is birth suppression, not a model of territorial fronts stopping at collisions, extinction, strategic responses, or resource exhaustion. Persistent and perfectly detectable signatures make seeing no signature at time T equivalent to no signal having arrived by T.

Let S_T be the accepted-process silence event. Conditional on S_T, let $Q>0$ be the delay from T to the first accepted signal and $H>0$ the delay to the first accepted arrival. On finite-contact events, $Q \le H$ and $H-Q$ is the warning interval following the first signal from *any* accepted source. Use H_0 for the free process's contact delay.

Main theorem

Theorem 1. Speed-tail transition and warning law

Assume the preceding model and

Equation E1.

$$F(z) \sim C\, z^{\alpha} (z \downarrow 0), C > 0, \alpha > 0.$$

Set $\rho = min(\alpha, d)$ and $p_T(h) = P(H \le h \mid S_T)$. Then, as $h \downarrow 0$ with T fixed,

Equation E2.

$$p_T(h) \sim \begin{cases} \dfrac{\lambda v_d c^d C d}{(d-\alpha)(\alpha+1)} T^{d-\alpha} h^{\alpha+1}, & 0 < \alpha < d, \\ \dfrac{\lambda v_d c^d C d}{d+1} h^{d+1} \log(T/h), & \alpha = d, \\ \dfrac{\lambda v_d c^d}{d+1} E\left[q^{-d}\right] h^{d+1}, & \alpha > d. \end{cases}$$

The negative moment in the last line is finite. Conditional on S_T and $H \le h$,

Equation E3.

$$\left(\frac{Q}{h}, \frac{H-Q}{h}, 1 - \frac{H}{h}\right) \Rightarrow Dirichlet(1, \rho, 1).$$

Equivalently, the pair

$$\left(\frac{H}{h}, \frac{H-Q}{H}\right) \Rightarrow (Z, U)$$

has independent limits $Z \sim Beta(\rho+1, 1)$ and $U \sim Beta(\rho, 1)$.

The Dirichlet notation means that the limiting density of the first two coordinates (x, y) is

Equation E4.

$$\rho(\rho+1) y^{\rho-1} 1\{x > 0, y > 0, x + y < 1\}.$$

This definition makes the joint claim explicit even without prior familiarity with the Dirichlet family.

For $d = 3$, the regular finite-moment regime has an h^4 early-contact probability and limiting warning-fraction CDF u^3. A tail $F(z) \sim C\, z$ instead gives an h^2 probability and a uniform warning fraction. A tail with $\alpha = 3$ retains the u^3 shape but changes the probability to order $h^4 \log(T/h)$. These statements are conditional on contact in a shrinking horizon; they do not say that contact is likely soon.

Proof: causal silence and the thin signal shell

Lemma 2. Causal silence identity

The accepted and free processes have the same event of no signal by T. In particular,

Equation E5.

$$P(S_T) = \exp\left(-\frac{\lambda v_d c^d T^{d+1}}{d+1}\right).$$

This probability does not depend on F.

Proof.

First, the accepted process is well-defined. All possible ancestors of a birth (x, t) lie in its backward light cone $0 \le s < t$, $|y - x| \le c(t - s)$. This cone has finite intensity and hence finitely many potential births almost surely. Sorting its points chronologically defines their acceptance; no infinite backward chain is possible. This construction is consistent over cones.

If an accepted birth i suppresses a potential birth j, then

$$|x_i - x_j| \le V_i(t_j - t_i).$$

By the triangle inequality and $V_i < c$,

$$D_i = t_i + |x_i|/c \le t_i + |x_i - x_j|/c + |x_j|/c \le t_j + |x_j|/c = D_j.$$

Consequently every rejected potential signal has an accepted signal arriving no later. A potential signal by T therefore entails an accepted signal by T. The converse is immediate. The cone $D \le T$ has intensity

$$\lambda \int_0^T v_d c^d (T - t)^d \, dt = \frac{\lambda v_d c^d T^{d+1}}{d+1},$$

and the Poisson void formula proves (E5).

End of proof.

The substance of this light-cone observation is already present in Olson's treatment of observability with multiple expansion strategies [1, Section IV.E]. It is included as a proof dependency, not claimed as a new principle.

Define sets in marked birth space by

$$B_h = \{T < D, A \le T + h\}, C_h = \{T < D \le T + h\}.$$

Since $D < A$, $B_h \subset C_h$. Conditional on S_T, the points outside the empty past light cone retain their original Poisson law. Let

$$\begin{aligned} b(h) &= \mu(B_h), \\ s(h) &= \mu(C_h) = \frac{\lambda \nu_d c^d}{d+1}\left[(T+h)^{d+1} - T^{d+1}\right) = O(h). \end{aligned}$$

In particular, $b(h) \le s(h) \to 0$.

Lemma 3. Quantified suppression bound

For every $h > 0$,

$$\begin{aligned} b(h)e^{-s(h)} &\le p_T(h) \le 1 - e^{-b(h)}, \\ 0 \le \left(1 - e^{-b(h)}\right) - p_T(h) &\le b(h)s(h). \end{aligned}$$

It follows that $p_T(h) \sim b(h)$ as $h \downarrow 0$. The conditional early-contact marked laws of the free and accepted processes differ by $O(h)$ in total variation.

Proof.

An accepted contact requires a potential point in B_h, proving the upper bound. If exactly one point lies in C_h and that point belongs to B_h, it cannot have been rejected: an accepted suppressor would have signal time no greater, would also belong to C_h under silence, and would be a second point. The unique point therefore causes contact. This event has probability $b(h)e^{-s(h)}$.

A disagreement about early contact or its responsible source is possible only on

$$E_h = \{N(B_h) \ge 1, N(C_h) \ge 2\}.$$

Poisson factorial moments give

$$P(E_h \mid S_T) \le E\left[N(B_h)\left(N(C_h) - 1\right) \mid S_T\right) = b(h)s(h).$$

This proves (E7). Normalizing the coupled marked subprobability laws by their early-contact probabilities gives, for example, the valid bound $2\,s(h)e^{s(h)}$ for their total-variation distance. The bound tends to zero at order h.

On the complement of E_h, an early contacting source is also the first source of any signal after T. The same estimate therefore permits replacing that source's own warning by the warning after the global first signal. No derivative of the error estimate is used.

End of proof.

The qualification in the last sentence is important: this coupling proves statements conditional on $H \le h$, not automatically pointwise statements conditional on $H = h$.

Proof: pair intensity and the exponent transition

For a potential source put $\delta = D - T > 0$ and $w = A - D > 0$. At fixed q,

$$r=|x)=\frac{c\,w}{q},\,t=T+\delta-\frac{w}{q}.$$

The radial volume element is $d\,v_d\,r^{d-1}\,d\,r$. The absolute Jacobian of $(\delta,w)\mapsto(t,r)$ is c/q. Thus the marked intensity is

$$\lambda\,d\,v_d\,c^d\,q^{-d}\,w^{d-1}\,d\,\delta\,d\,w\,d\,F(q).$$

The lower bound $t\geq 0$ is exactly $q\geq w/(T+\delta)$. Introduce the truncated negative moment

$$J(a)=\int_{[a,\infty)} q^{-d}\,d\,F(q),\,a>0.$$

After integrating out the speed mark, the exact signal/warning intensity is

Equation E8.

$$\lambda\,d\,v_d\,c^d\,w^{d-1}\,J\left(\frac{w}{T+\delta}\right)d\,\delta\,d\,w.$$

Early-contact points occupy the triangle $\delta>0$, $w>0$, $\delta+w\leq h$.

The only asymptotic integral needed

Stieltjes integration by parts and (E1) imply

Equation E9.

$$J(a)\sim\begin{cases}\dfrac{C\,\alpha}{d-\alpha}\,a^{\alpha-d}, & 0<\alpha<d,\\ C\,d\log(1/a), & \alpha=d,\\ E\left[q^{-d}\right), & \alpha>d.\end{cases}$$

For clarity, integrate first on $(a,a_0]$, with $a_0>0$ fixed and small:

$$\int_{(a,a_0]} q^{-d}\,d\,F(q)=a_0^{-d}\,F(a_0)-a^{-d}\,F(a)+d\int_a^{a_0} q^{-d-1}\,F(q)\,d\,q.$$

When $\alpha<d$, the last two terms have leading coefficient $-C+d\,C/(d-\alpha)=C\,\alpha/(d-\alpha)$. When $\alpha=d$, the integral is $C\,d\log(1/a)$ to first order. When $\alpha>d$, the near-zero negative moment converges; the part with $q\geq a_0$ is bounded. If F has atoms, their endpoint contribution is negligible: $F(a-)\sim C\,a^{\alpha}$ follows by squeezing between $F((1-\epsilon)a)$ and $F(a)$ and then sending $\epsilon\downarrow 0$.

For $0<\alpha<d$, substituting (E9) in (E8) gives a limiting density proportional to

$$T^{d-\alpha}\,w^{\alpha-1}.$$

Its triangle integral uses

$$\int_0^h \int_0^{h-\delta} w^{\alpha-1}\, dw\, d\delta = \frac{h^{\alpha+1}}{\alpha(\alpha+1)}.$$

This yields

$$b(h) \sim \frac{\lambda v_d c^d C d}{(d-\alpha)(\alpha+1)} T^{d-\alpha} h^{\alpha+1}.$$

For $\alpha > d$, the same integral with power $d-1$ yields $b(h) \sim \lambda v_d c^d E\left[q^{-d}\right] h^{d+1}/(d+1)$. At $\alpha = d$, it is multiplied by $C d \log(T/h)$, giving the middle line of (E2).

Here limit-integral interchange is justified explicitly. Scale $\delta = h x$, $w = h y$ on the fixed unit triangle. In the first regime, (E9) bounds the normalized integrand by a constant times $y^{\alpha-1}$ for all sufficiently small h. In the finite-moment regime it is bounded by a constant times y^{d-1}. At criticality the normalized logarithm is bounded by a constant times $1+|\log y)$, which is integrable against y^{d-1}. These are integrable majorants in every case since $\alpha, d > 0$.

By Lemma 3, these asymptotics for $b(h)$ are also the accepted-process probability asymptotics. This proves (E2).

The joint limit

After the same scaling and normalization, the density of a potential early-contact point tends in L^1 to (E4). Given at least one point in B_h, there is exactly one with probability tending to one, since $b(h) \to 0$. Its mark law therefore has the same limit as the first free arrival. Lemma 3 transfers that law to the accepted process and permits use of its global first-signal time Q.

Finally set $z = x + y$ and $u = y/(x+y)$. The Jacobian is z, so

$$\rho(\rho+1)(zu)^{\rho-1} z = \left[(\rho+1) z^{\rho}\right]\left[\rho u^{\rho-1}\right], 0 < z, u < 1.$$

This factors into the claimed independent beta densities, proving all of Theorem 1. ▫

Exact formulas and reproducible checks

Finite-age kernel

For fixed q, radial integration gives

Equation E10.

$$b(h) = \frac{\lambda v_d c^d}{d+1} \int K_T(q,h)\, dF(q),$$

where

$$K_T(q,h) = \begin{cases} h^{d+1}/q^d, & q \geq h/T, \\ (T+h)^{d+1}/(1+q)^d - T^{d+1}, & 0 < q < h/T. \end{cases}$$

Both branches agree at $q = h/T$. To verify the formula directly, let $z = r/c$ and integrate the admissible birth-time length

$$\lambda d v_d c^d \int_0^\infty z^{d-1} [T + h - (1+q)z - max(0, T - z)]_+ dz.$$

Splitting at $z = T$ gives (E10). The exact free-process probability is $1 - e^{-b(h)}$. Equation (E6), not this exact exponential, applies to the heterogeneous accepted process.

Discarding the second branch incorrectly replaces the entire kernel by $h^{d+1} E[q^{-d}]$. That moment is infinite for $\alpha \le d$, but the correct finite-age mean is finite and bounded by $s(h)$. The apparent divergence is an invalid approximation, not an instantaneous-contact effect.

Exact conditional warning at a specified free arrival

Let W_* denote the warning from the *particular free source* that arrives first. For $0 \le u \le 1$, define

$$a = \frac{uh}{T + (1-u)h}$$

and

$$L(h,u) = \lambda v_d c^d [\quad (T+h)^d \int_{(0,a]} (1+q)^{-d} dF(q)$$

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Set $L(h,0) = 0$. Then

$$b'(h) = L(h,1), P(W_*/H_0 \le u \mid H_0 = h, S_T) = \frac{L(h,u)}{L(h,1)}.$$

The void factor for all earlier free arrivals cancels in this conditional mark law. This finite-h identity is source-specific. The global warning claim in Theorem 1 follows from the separate asymptotic coupling.

If $q \ge q_{min} > 0$, then for $h < T\, q_{min}$ the first branch of (E10) is exact and the last conditional CDF is exactly u^d.

Two closed-form examples

Use dimensionless units with $d = 3$, $T = c = 1$, $\lambda v_3 = 1$, and $F(q) = q^\alpha$ for $0 < q < 1$. For $0 < h < 1$:

α	$b(h)$	$P(W_*/H_0 \le u \mid H_0 = h)$
1	$\frac{3}{4}h^2 + \frac{1}{2}h^3$	$u[1 + h(1-u)]$
2	$h^3 - \frac{1}{4}h^4$	$u^2 \frac{3 + h(3 - 4u)}{3 - h}$

These are identities in the free process. Their leading probabilities and integrated early-contact warning limits transfer to the accepted process.

For $\alpha = 3$,

$$b(h) = \frac{3}{4} h^4 \log(1/h) + \frac{7}{16} h^4 + O(h^5).$$

Thus convergence of the critical relative asymptotic is only logarithmic.

High-precision computations

The supplied script verify_contact_asymptotics.py uses Python's standard-library decimal at 100-digit precision. It integrates the convergent binomial series for $(1+q)^{-3}$ and tests $\alpha \in \{0.5, 1, 2, 3, 4, 6\}$ and $h \in \{10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-12}\}$. At $h = 10^{-8}$:

α	Exact $b(h)$/leading asymptotic	Maximum sampled warning-CDF error
0.5	1.00000001000000004	2.9463×10^{-9}
1	1.00000000666666667	2.5000×10^{-9}
2	0.99999999750000000	1.6667×10^{-9}
3	1.03166730608068573	4.6200×10^{-3}
4	0.99999999400000001	5.4675×10^{-10}
6	$1 - O(10^{-25})$	9.8780×10^{-26}

The CDF check uses $u = 0.1, 0.5, 0.9$ and compares with $u^{\min(\alpha,3)}$. A five-point numerical derivative independently checks $b'(h) = L(h,1)$, with relative error below 4.1×10^{-34} on this grid. These are finite consistency checks; the proof establishes the general result.

Direct simulation with birth suppression

The separate script simulate_suppressed_contacts.py samples the actual spatial process with $\alpha = 1$, $d = 3$, $c = T = 1$, and $\lambda v_3 = 1$. It conditions on a free early contact by drawing independent Poisson restrictions to B_h and $C_h \setminus B_h$, with the former count conditioned to be positive. It samples positions and times continuously and then applies the chronological accepted-birth rule. Every relevant suppressor is inside C_h, so no omitted exterior region or spatial/time grid approximation enters this construction.

For $h = 0.02$, 100,000 trials with seed 20261101 gave the following means in the accepted process:

Quantity	Simulated mean	Limiting mean
H/h	0.667229	$2/3$
$(H-Q)/H$	0.498541	$1/2$
Q/h	0.334453	$1/3$
$(H-Q)/h$	0.332776	$1/3$
$1 - H/h$	0.332771	$1/3$

There were no suppressed births in this small-horizon run, so it alone cannot test the suppression branch. The global first signal and the contacting source's own signal differed in 0.007 of trials, confirming that the finite-horizon warning distinction is substantive. Standard errors of the horizon-scaled means were about 0.00074.

To exercise suppression, a separate finite-horizon check used $h = 0.5$, 50,000 trials, and seed 20261102. It produced 416 rejected births, including 107 in B_h; five contact winners changed, and nine free early contacts were eliminated. First detection remained unchanged in every trial. This check demonstrates why a finite-horizon equality between free and accepted contacts would be false. It does not establish the asymptotic theorem, which rests on the preceding proof. Both full result files and the sampling formulas are supplied with the code.

A further connection: memory of the birth history

The exponent transition also changes which part of the birth history determines the leading coefficient. Replace $\lambda\, d\, x\, d\, t$ by $\lambda(t)\, d\, x\, d\, t$, where $\lambda(t)$ is bounded and continuous on $[0, T+\epsilon)$, nonnegative, and $\lambda(T) > 0$. Keep independent speed marks with (E1).

For $0 < \alpha < d$, the same probability and warning proof gives

Equation E11.

$$p_T(h) \sim \frac{d\, v_d\, c^d\, C}{\alpha + 1} h^{\alpha+1} \int_0^T \lambda(t)(T-t)^{d-\alpha-1}\, d\, t\,.$$

For $\alpha > d$, the leading coefficient is instead $\lambda(T)\, v_d\, c^d\, E\left[q^{-d}\right)/(d+1)$; at $\alpha = d$, it is $\lambda(T)\, v_d\, c^d\, C\, d/(d+1)$ multiplying $h^{d+1}\log(T/h)$. The Dirichlet limit remains unchanged.

Here is a direct derivation of the historical coefficient. At candidate arrival delay h, set $r = c(T+h-t)/(1+q)$. The exact free arrival intensity is

$$\eta_T(h) = d\, v_d\, c^d \int_0^{T+h} \lambda(t)(T+h-t)^{d-1} \int \frac{1\{q(T-t) < h\}}{(1+q)^d}\, d\, F(q)\, d\, t\,.$$

For fixed $t < T$, the inner integral is asymptotic to $C\, h^{\alpha}(T-t)^{-\alpha}$. The singularity of the resulting integrand at $t = T$ is integrable precisely when $\alpha < d$. Splitting a small neighborhood of T, using boundedness of λ, and then applying dominated convergence yields

$$\eta_T(h) \sim d\, v_d\, c^d\, C\, h^{\alpha} \int_0^T \lambda(t)(T-t)^{d-\alpha-1}\, d\, t\,.$$

Integration in h proves (E11). To make the domination explicit, put $s = T - t$. The part with $0 \le s \le h$ and the future-birth part are $O(h^d) = o(h^{\alpha})$. On $s \ge h$, the bounds $F(h/s) \le K(h/s)^{\alpha}$ and $(s+h)^{d-1} \le 2^{d-1} s^{d-1}$ give an integrable majorant proportional to $s^{d-\alpha-1}$ after division by h^{α}.

For the other regimes split birth times at $T - \epsilon$. The far-past contribution to the cumulative mean is $O(h^{\alpha+1})$, and hence is negligible against h^{d+1} when $\alpha > d$, or against $h^{d+1}\log(T/h)$ when $\alpha = d$. On the recent interval

continuity sandwiches $\lambda(t)$ between arbitrarily close constants around $\lambda(T)$. Applying the proved constant-rate formulas with history length ϵ and then shrinking ϵ yields the asserted coefficients.

For completeness the varying-rate joint signal/warning intensity is

$$\kappa(\delta, w) = d\, v_d\, c^d\, w^{d-1} \int_{[w/(T+\delta),\infty)} \lambda(T+\delta-w/q)\, q^{-d}\, dF(q).$$

If $0<\alpha<d$, rescale $q = w\,u$. The measures $w^{-\alpha}\, dF(w\,u)$ converge vaguely on positive compact intervals to $C\,\alpha\, u^{\alpha-1}\, du$. Their weighted tails satisfy

$$\int_M^\infty u^{-d}\, w^{-\alpha}\, dF(w\,u) = w^{d-\alpha}\, J(w\,M) \le K\, M^{\alpha-d}$$

for fixed sufficiently large M and all sufficiently small w. Boundedness and continuity of λ then give, as $(\delta, w) \to (0,0)$ through positive values,

$$\frac{\kappa(\delta, w)}{w^{\alpha-1}} \to d\, v_d\, c^d\, C\,\alpha \int_0^T \lambda(T-s)\, s^{d-\alpha-1}\, ds.$$

The bound $\kappa(\delta, w) \le K'\, w^{\alpha-1}$ on the shrinking triangle proves the same Dirichlet law without assuming that F has a density. In the high and critical regimes the preceding recent-history sandwiches also apply to the joint intensity. Finally the shell coupling continues to apply because its intensity remains $O(h)$.

Up to the conventional factor $\Gamma(d-\alpha)$, the integral in (E11) is a fractional integral of the past birth rate. Thus the speed tail changes both the exponent and whether the leading coefficient depends on the entire past or only the present local rate. Fractional integrals and regular-variation mechanisms themselves are established mathematics.

Critical examination and counterexamples to overclaims

The theorem does not identify the actual universe

Poisson attempts, homogeneity, a continuum of birthplaces, eternal spherical expansion, a known finite starting time, and perfect immediate signatures are assumptions. The power-law tail near light speed is an additional assumption. Neither it nor its exponent has been measured for extraterrestrial civilizations. This theorem therefore does not favor a physical explanation such as rarity, concealment, self-destruction, or near-light-speed expansion.

The silence event here is stronger than a finite collection of unsuccessful astronomical searches. Incomplete detection, emission delay, finite signal lifetimes, beaming, or frequency restrictions change the event being conditioned on. A fixed point in space is also different from an observer selected by an anthropic sampling rule. Conditioning on being a surviving civilization can introduce further speed dependence.

Discrete sources can change the exponent

Suppose instead that births occur at finitely many fixed locations at radii $r_i > 0$, with rates λ_i. If speeds are bounded above by $v_{max} < c$, ideal silence implies a strictly positive contact gap:

$$H \geq \min_i r_i \left(\frac{1}{v_{max}} - \frac{1}{c} \right) > 0 .$$

Indeed every later arrival has $A - T = (D - T) + (A - D)$, where the first term is positive and the second is at least the displayed bound.

If speeds approach c with $F(z) \sim C z^{\alpha}$ and $0 < r_i < c T$, then the contribution of site i is instead

$$\lambda_i \int \left(h - \frac{q r_i}{c} \right)_+ d F(q) \sim \frac{\lambda_i C}{\alpha + 1} \left(\frac{c}{r_i} \right)^{\alpha} h^{\alpha + 1} .$$

There is no local continuum contribution forcing saturation at exponent $d + 1$. This is a concrete counterexample to interpreting the theorem's dimension threshold as universal for any galaxy model. Spatial discreteness, finite survey scales, and cosmological geometry need separate treatment.

Other boundaries of the result

- An atom $p > 0$ at $V = c$ is excluded. It produces a leading contact probability $\lambda v_d c^d p T^d h$ and zero leading warning, a different limit.
- Finite-h identities for free fronts cannot simply be copied into the accepted model when speeds vary. The proved transfer is asymptotic, with the explicit error (E7).
- The warning supplied by the contacting source and warning following the first signal from any source differ at finite horizons. Their equivalence here is a conditional small-horizon result.
- The result is not uniform over T, λ, or F. Sending $T \to \infty$, $\lambda \to \infty$, or $\alpha \to d$ together with $h \to 0$ requires new arguments. At $\alpha = d$, logarithmically slow convergence is substantial even for tiny h.
- Strategic behavior, collision shielding, finite expansion lifetimes, and correlations between speed and birth environment are outside the proved model.

Originality assessment and significance

What was checked

The literature search examined expanding-civilization models, Johnson–Mehl–Avrami–Kolmogorov growth, random growth velocities, phantom overgrowth, first-contact and warning distributions, causal thinning, and regular variation. It included the following specific primary work:

1. Olson's 2015 paper [1], including its discussion of fast phantom bubbles escaping slow real bubbles and its light-cone observability formula. This is direct prior art for the model and for Lemma 2 in substance.
2. Olson's 2018 and 2020 papers [2, 3], which give expansion and visibility predictions and anthropic conditioning under specified assumptions. Poisson cone methods and the inference problem are therefore established.
3. Bhattacharjee, Molchanov, and Turin [4], who study random-speed Poisson birth-growth and explicitly distinguish free growth from physically suppressed nucleation. Their principal theorem is a central limit theorem for exposed-point statistics.

4. Birnie and Weinberg [5], whose abstract and institutional record describe one-dimensional continuous nucleation with distributions of velocities, shielding, and phantom effects. Its full theorem set was not inspected here, so it remains a possible source of additional overlap.
5. Denisov and Zwart [6] on product-tail asymptotics, and Pakkanen, Sottinen, and Yazigi [7, Lemma 3.4] on regular-variation convolutions. Competition between exponents, borderline logarithms, and beta-function limits are established analytic mechanisms.

No exact prior statement of Theorem 1, with the global warning law and quantified suppression transfer, was located. This is limited negative search evidence. It is not a proof of historical originality, and terminology in another field may conceal an equivalent theorem.

What is and is not claimed

Claim	Assessment
Poisson void probabilities and causal cone methods	Known tools
Random-speed nucleation and phantom-birth difficulty	Known model and issue
Regular-variation exponent competition and beta limits	Known mechanisms
The conditional three-regime formula, Dirichlet law, and explicit suppression coupling assembled here	Proved here; exact historical novelty unresolved
An empirically established explanation of the Fermi paradox	Not achieved

The mathematical interest lies in identifying the correct finite-age truncation, deriving a joint conditional law with explicit constants, and showing that a causal interaction affecting the full process does not alter the leading conditional short-time geometry. The theorem also gives a constructive identifiability example: for fixed λ, c, T, all speed laws have the same silence likelihood (E5), while they can produce different contact exponents and warning laws. That is an obstruction to selecting among these futures from this particular null datum alone.

Final verdict. A model theorem with a complete analytic proof and reproducible computational checks was obtained. The specific combined result is a candidate original refinement, with novelty unverified and no claim of publication-level priority. The empirical Fermi paradox remains unresolved.

References

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